

Project Assignment 2 Report: Medicaid

Stakeholder question:

- **Problem/Question:** “How has the claim quantity and Medicaid payout of medical services fluctuated throughout COVID? Which services went up, which went down, which were unaffected? Which services were permanently affected after the pandemic?”
- **Answer:** Based on the sample, 2020 was seemingly the year with the least amount of claims, with 2021 and 2022 being the year with the most.

Roles:

- **Team Members:**
 - Marius – Product Manager (Sprint 2)
 - Dylan – Data Engineer (Sprint 2)
 - Justin – Data Engineer (Sprint 2)
 - Devin - Data Engineer (Sprint 2)
 - Rupashi – Data Storyteller (Sprint 2) author

Profiling summary:

Sample_1000.csv:

- **Size (bytes):** 49850
- **Rows:** 1007

Frequency Tables:

Claim years: *this gave a pretty surprising result with 2020 at the bottom and the years following being at the top. This may just be because of the small sample size.*

- 168 of 2022
- 159 of 2023
- 148 of 2019
- 146 of 2021
- 142 of 2024
- 123 of 2020
- 121 of 2018

HCPCS codes: *this gave a pretty expected result, office visits are the most frequent*

- 72 of 99213: Established patient office visit, 20-29 minutes
- 62 of 99214: Established patient office visit, 30-39 minutes
- 22 of 99285: Emergency department visit, high complexity
- 22 of 99284: Emergency department visit, moderate-to-high complexity
- 16 of T1015: Clinic visit/encounter, all-inclusive
- 16 of 99283: Emergency department visit, low-to-moderate complexity
- 12 of 85025: Complete blood count (CBC) with automated differential WBC count
- 11 of D0220: Intraoral periapical first radiographic image (Dental X-ray)
- 11 of 90837: Psychotherapy, 60 minutes
- 10 of D1120: Child prophylaxis cleaning (Dental cleaning)
- 10 of D0120: Periodic oral evaluation, established patient (Dental exam)
- 10 of 36415: Collection of venous blood by venipuncture (Routine blood draw)

Top-10 servicing providers:

Simply wasn't enough data in the sample to get any meaningful results

- 2 of A565813600
- 2 of 1932145778
- 2 of 1750704680
- 1 of M961997600
- 1 of A962823100
- 1 of A599428400
- 1 of A302139400
- 1 of 1992979454
- 1 of 1992893416
- 1 of 1992783286

Filter recent summary:

It is really good to know that even with such a small sample size, we're still able to get results for 15000 patients

- Matching records: 301
- Total patients: 15769
- Total claim lines: 22603

Data quality:

Some of the fields of the Medicaid data-set were empty, specifically the clinic and provider NPI fields. I'm not sure why this is done (perhaps anonymity?), but we had to clear those in the sample. To do this, when selecting lines for the sample, we specifically ensure that those fields were not empty. In general this sample is fine, but not all analysis is realistically possible with such a small size. The largest bottleneck in our .sh script is iterating through the original medicaid data-set to generate the sample. This takes about 5 minutes, but the rest of the analysis takes just a second. So we would benefit from increasing the sample size to a more meaningful size such as 10,000.

Links:

Github: https://github.com/sjsu-cs131-f26/team4_medicaid_analysis/tree/main

Pull requests:

- https://github.com/sjsu-cs131-f26/team4_medicaid_analysis/pull/1
- https://github.com/sjsu-cs131-f26/team4_medicaid_analysis/pull/2
- https://github.com/sjsu-cs131-f26/team4_medicaid_analysis/pull/3
- https://github.com/sjsu-cs131-f26/team4_medicaid_analysis/pull/4

Board: <https://trello.com/b/k71Nv6qN/cs131team4>

Original data-set: <https://opendata.hhs.gov/datasets/medicaid-provider-spending>

AI use disclosure:

AI was of limited use to help resolve issues. Specifically for using git and for understanding why some shell commands produced unexpected results. We also used it to help us generate the timer, since this was not something we could've figured out ourselves. We did not use AI to generate any of the text itself, it is all hand written.

Stakeholder update:

Our next step is to increase the sample size and run some more tests. We are going to start looking at the frequency and payout of services between the ranges of 2018 to 2020, 2020 to 2022, 2022 to 2024. We're hoping with this data we'll get a rough idea of some trends we can further elaborate on in the future. We're also thinking about if there's any useful data we can extract from the clinic and provider NPI's. So far we have some ideas, but nothing concrete yet.